Stock-Bond Correlation Regimes Across Crisis Types

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Overview

For most of the past two decades, bonds served as a reliable hedge. During equity drawdowns, U.S. Treasuries typically rallied, allowing a simple 60/40 portfolio to dampen volatility and preserve capital. This relationship broke down in 2022, when inflation surged, the Federal Reserve aggressively tightened policy, and both stocks and bonds declined simultaneously, producing one of the worst years on record for the 60/40 framework.

This project investigates whether **the type of macroeconomic shock** explains this breakdown. Specifically, I test whether:

- Growth-driven crises (e.g., 2008, 2020) lead to negative stock-bond correlation
- Inflation-driven crises (e.g., 2022) produce positive or weakened correlation

Using 21 years of daily data, I find strong evidence that crisis type drives correlation regimes, and that volatility and the yield curve together, rather than either in isolation, are needed to distinguish the regimes predictively.

Data & Methods

Data

The dataset includes daily observations from January 2005 to April 2026 for:

- SPY (equities), TLT (long-duration Treasuries)
- 11 SPDR sector ETFs
- VIX (market-implied volatility)
- 10Y-2Y Treasury yield spread (FRED)

This period captures four major stress events: the 2008 financial crisis, the 2020 COVID crash, the 2022 inflation shock, and the April 2025 tariff-driven drawdown.

Methods

- A GARCH(1,1) model to estimate varying volatility and define volatility regimes.
- A DCC-GARCH model (implemented from scratch via maximum likelihood estimation) was used to generate a dynamic time series of stock-bond correlation.
- Rolling 60-day correlations and sector heatmaps provided descriptive validation.
- A lead-lag analysis tested whether the yield curve predicts future correlation regimes.

Validation

- GARCH volatility aligns closely with the VIX ($\rho = 0.881$).
- DCC correlation and rolling correlation are highly consistent ($\rho = 0.879$).

Key Findings

1. **The stock-bond correlation flipped in 2022 and has not reverted.**

From 2005, through 2021, the average SPY-TLT 60-day correlation was -0.41. The correlation deepened further during the growth-driven crises, reaching -0.46 during the 2008 financial crisis and -0.50 during the 2020 COVID crash, consistent with classical flight-to-safety dynamics. In 2022, the correlation jumped to +0.09, and during the April 2025 tariff drawdown it registered +0.16. Three years into the new regime, including an out-of-sample stress event, the hedge has not re-established. The DCC-GARCH persistence parameter (b = 0.936) supports this interpretation: the model implies that once correlations enter a new regime they tend to persist rather than rapidly mean-revert.

The 2025 value requires a minor caveat. Decomposing the tariff window into sub-phases reveals that the correlation during the acute crash was near zero, while the post-event recovery produced strongly positive correlation as equities rebounded and Treasuries sold off. The headline +0.16 therefore reflects the recovery phase more than the crash itself, but it remains well above the pre-2022 baseline and is consistent with the new regime.

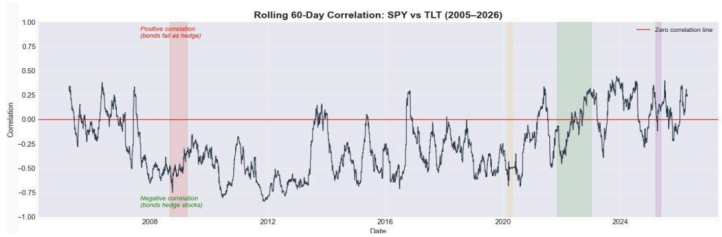

Figure 1. Rolling 60-day correlation between SPY and TLT. Negative for most of 2005-2021, then flips positive in 2022 and remains elevated through 2025.

STOCK-BOND CORRELATION BY CRISIS PERIOD

-0.46	-0.50	+0.09	+0.16
2008 Crisis	2020 COVID	2022 Inflation	2025 Tariff

Growth-driven crises (blue) preserve the bond hedge. Inflation-related crises (red) weaken or reverse it.

2. **Volatility identifies stress, but does not predict correlation direction.**

To test whether volatility alone can identify stock-bond regime shifts, I classified each trading day into a volatility regime using the estimated GARCH conditional volatility, then examined the stock-bond correlation within the high-volatility bucket across different eras. The same high-volatility regime produced an average correlation of -0.496 before 2021 (n = 1, 302 days) and +0.050 during 2021-2022 (n = 288 days). A Welch's t-test on the underlying daily co-movement distributions confirms the shift is highly statistically significant ($p < 0.001$).

Volatility, the most common regime signal, indicates that stress is occurring, but not whether bonds will hedge or fail. The type of shock determines correlation direction, and volatility alone cannot identify it.

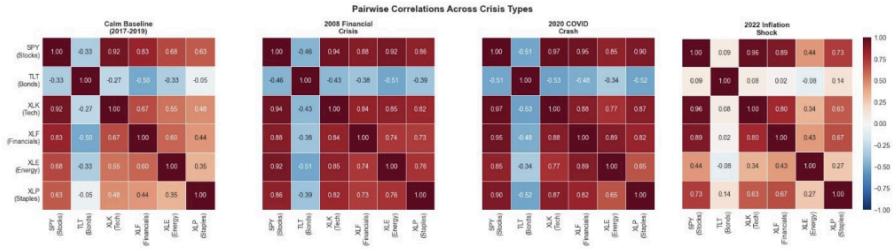

Figure 2. Correlation heatmaps by regime. The TLT row illustrates the regime distinction: deep blue in 2008 and 2020 (bonds hedging equities) and near zero in 2022 (hedge absent). Equity-to-equity correlations tighten in every crisis; only the bond relationship differs across crisis types.

3. **The yield curve's predictive power has weakened substantially post-2022.**

A lead-lag analysis between the 10Y-2Y Treasury spread and the DCC conditional correlation identifies a full-sample peak of -0.52 at a three-month horizon, a meaningful and economically interpretable signal. In the post-2022 sample, the three-month peak is -0.18, approximately half the pre-2022 strength. The stronger full-sample estimate is partly attributable to the joint movement of both series through the 2022 regime transition rather than a sustained predictive relationship. The yield curve was a useful leading indicator in the pre-2022 environment, but has become substantially less informative in the current regime.

Strategic Implications

Two implications follow from the results. Both are stated at the level of portfolio-risk rather than as specific trade recommendations.

- **Single-signal regime detection is insufficient.** Finding 2 demonstrates that volatility cannot distinguis a growth-driven crisis (where bonds hedge) from an inflation-driven crisis (where they do not). A framework that treats elevated volatility as a uniform buy-bonds signal will fail systematically in inflation regimes. Practitioners relying on volatility-based regime detection should supplement it with an independent read on the inflation environment.
- **The yield curve retains value but requires an era-aware caveat.** The weakening of the yield curve signal post-2022 (Finding 3) implies that practitioners using curve-based regime forecasting should not treat a current-era inversion as equivalent in information content to a pre-2022 inversion.

The April 2025 tariff drawdown highlights these implications. Naive volatility- or yield-curve-based frameworks would yield ambiguous classifications, while the observed stock-bond correlation (+0.16) places it in the post-2022 inflation-correlation regime, indicating that tariff-driven inflation effects dominate growth effects over a multi-week horizon.

Limitations

Several caveats apply. The post-2022 sample (~1,000 trading days) limits statistical power for regime-specific claims. The lead-lag analysis is technically in-sample, using full-sample parameter estimation. An initial sector-dispersion finding from the 2022 inflation shock was discarded after a matched-window check showed it was driven by unequal window lengths, not true dynamics. The April 2025 tariff event offers only a brief out-of-sample window and should not be overinterpreted as evidence of regime persistence.

References

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